

Michael A.

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PROFESSIONAL SUMMARY

Senior Data Engineer with 10+ years delivering scalable cloud-native data solutions in insurance, healthcare, and telecom domains. Proven expertise in architecting high-volume ETL/ELT pipelines, real-time streaming, cloud migrations (AWS/Azure), and data governance to support analytics and reporting. Skilled in Snowflake data warehousing with Medallion architecture, Kafka streaming, and orchestration with Azure Data Factory and Apache Airflow. Strong collaborator and mentor with light ML integration experience supporting AI-driven analytics.

TECHNICAL SKILLS

- **Cloud Platforms:** AWS, S3, Redshift, Glue, Lambda, Azure, Data Factory, Databricks, Data Lake
- **Data Engineering:** Python, PySpark, SQL, Apache Kafka, Spark Streaming, Apache Airflow, DBT, Terraform, Java
- **Data Warehousing & Modeling:** Snowflake, Redshift, Star/Snowflake Schema, Medallion Architecture, Data Vault 2.0
- **Orchestration:** Azure Data Factory, Apache Airflow, Microsoft Fabric basics, Prefect
- **Databases:** PostgreSQL, SQL Server, Oracle, MongoDB, Cosmos DB
- **DevOps & Automation:** Jenkins, Kubernetes, Docker, Azure DevOps, GitHub Actions
- **Monitoring & Governance:** Tableau, Prometheus, Grafana, Colibra

PROFESSIONAL EXPERIENCE

GEICO

Senior Data Engineer

Apr 2022 - Present

Chevy Chase, MD

- Led cloud modernization initiative migrating on-prem insurance data platforms to AWS and Azure, improving ETL pipeline latency by 35%.
- Designed and implemented a scalable Snowflake data warehouse using the Medallion architecture (bronze, silver, gold layers), enabling underwriting, claims, and customer-analytics teams to access curated data faster and reducing report generation time
- Developed real-time streaming pipelines using Apache Kafka and Spark Streaming for RTD event processing, delivering near-real-time insights to downstream analytics
- Built reusable, modular ETL pipelines in Python and PySpark to automate data ingestion from policy management, claims, and telematics systems, cutting manual data-load effort and improving pipeline reliability
- Established data governance policies and automated quality checks to ensure compliance with insurance regulations and data accuracy
- Implemented REST APIs with Spring Boot to enable secure data sharing between third-party vendors and internal applications, improving data exchange speed and reliability
- Mentored junior engineers and promoted best practices in cloud data engineering using Python and Snowflake, leading cross-team technical workshops that raised team productivity and code quality
- Integrated lightweight AI-driven data summarization tools using embeddings, Vector Semantic Search, and LangChain to improve the efficiency of processing sales call notes.

Travelers Insurance

Senior Data Engineer

Mar 2019 - Apr 2022

Hunt Valley, MD

- Architected and optimized ETL pipelines processing large volumes of property, auto, and special lines insurance data for enterprise-wide analytics and reporting.
- Designed and implemented CDC and incremental data ingestion workflows using Kafka and Spark, enabling near real-time updates for customer dashboards and actuarial models.
- Automated cloud infrastructure provisioning across Azure and AWS with Terraform, accelerating deployments and improving environment reliability.
- Developed Star and Snowflake schema data models in Snowflake and Redshift aligned with insurance business domains such as claims, policies, and billing.
- Partnered with data science teams to deliver clean, integrated datasets supporting predictive analytics and risk scoring models.
- Implemented data governance frameworks with Apache Airflow and AWS to improve data quality, strengthen security, and meet insurance regulatory compliance
- Coordinated offshore and onshore teams to maintain smooth data ingestion, pipeline monitoring, and SLA adherence.
- Built CI/CD pipelines using Jenkins and Azure DevOps to automate testing and deployment of data workflows.

AT&T

Data Engineer

Dec 2016 - Mar 2019

Dallas, TX

- Designed scalable data pipelines for retail telecom data (customer usage, billing, service events) using Azure Data Factory and PySpark, enabling faster data ingestion for downstream analytics
- Built batch and streaming ETL pipelines in Azure Data Lake and Snowflake using PySpark and scheduled them with Apache Airflow, which cut reporting latency and supported real-time analytics
- Implemented monitoring and alerting mechanisms to track pipeline health, reducing data latency and failure rates.

- Collaborated with analytics and ML teams to deliver data marts enabling churn prediction and personalized marketing analytics.
- Developed automated data validation routines with dbt and Great Expectations, applying Data Vault 2.0 standards, which improved data quality and provided clear lineage for downstream users
- Led cloud migration efforts from legacy on-prem systems to cloud platforms, improving scalability and reducing operational costs.
- Implemented Azure DevOps pipelines for source control, continuous integration/deployment, and sprint planning, which shortened release cycles and improved delivery reliability for data-engineering projects
- Managed stakeholder communication to ensure alignment on data availability and reporting requirements.

INSA

Software Engineer

Feb 2012 - May 2016

Addis Ababa, Ethiopia

- Developed backend systems and APIs for health-insurance claims processing using Java, Spring Boot, and Oracle, enabling faster claim submission and validation
- Optimized database performance with complex SQL queries and PL/SQL stored procedures to support high-volume transactions.
- Integrated SOAP and REST web services enabling interoperability between healthcare providers and insurance platforms.
- Collaborated in Agile teams delivering software features aligned with healthcare regulatory compliance.
- Assisted the data team in automating ETL workflows for claims and policy data reconciliation using Azure Data Factory and Apache Airflow, which cut manual processing time and improved data accuracy

EDUCATION

University of Calabria

Master's, Computer Engineering

2014

Italy

University of Gondar

Bachelor's, Computer Science

2010

Ethiopia